

## Fateme GHAYEM

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| 2023–Now                | Postdoctoral research fellow<br>TANGRAM team, Inria, Nancy, France                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| CONTACT<br>INFORMATION  | Email: <a href="mailto:fatemeh.ghayem@inria.fr">fatemeh.ghayem@inria.fr</a><br><a href="mailto:fateme.ghayem@gmail.com">fateme.ghayem@gmail.com</a><br>Homepage: <a href="https://ghayem.github.io">https://ghayem.github.io</a><br>ORCID: <a href="https://orcid.org/0000-0001-5779-6685">https://orcid.org/0000-0001-5779-6685</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EDUCATION               | <b>Université Grenoble Alpes</b> , Grenoble, France<br><br>Ph.D. in Signal, Image, Parole, and Télécoms, <b>GIPSA-lab</b> , Oct 2017 – Nov 2020<br>- Thesis topic: <i>Optimal sensor placement for source extraction</i><br>- Advisor: Prof. Christian JUTTEN, Dr. Bertrand RIVET<br><br><b>Sharif University of Technology</b> , Tehran, Iran<br><br>M.Sc., <b>Electrical Engineering</b> , Sept 2013 – Sept 2015 GPA: 17.06/20<br>- Thesis topic: <i>MR image reconstruction from highly partial Fourier samples</i><br>- Advisor: Prof. Farokh Marvasti<br><br><b>Shiraz University</b> , Shiraz, Iran<br><br>B.Sc., <b>Electrical Engineering</b> , Sept 2009 – Sept 2013 GPA: 17.85/20<br><br><b>National Organization for Development of Exceptional Talents</b> , Shiraz, Iran<br><br>Diploma, <b>Mathematics and Physics</b> , Sept 2005 – Sept 2009 GPA: 19.60/20 |
| RESEARCH<br>INTERESTS   | <ul style="list-style-type: none"><li>– Image/text multi-modal foundation models for brain study</li><li>– Brain meta-analysis with large language models (LLM)</li><li>– Independent component/vector analysis (ICA/IVA) for multi-subject fMRI study</li><li>– Aneurysm detection with deep learning</li><li>– Optimal sensor placement for source extraction</li><li>– Machine learning and statistical signal processing in medical applications</li><li>– Bayesian modeling, numerical optimization, and dictionary learning</li></ul>                                                                                                                                                                                                                                                                                                                                |
| RESEARCH<br>EXPERIENCES | <ul style="list-style-type: none"><li>• <b>Postdoctoral researcher</b> (2025–Now), TANGRAM, Inria Nancy, France.<ul style="list-style-type: none"><li>- Advisor: Dr. Erwan Kerrien</li><li>- Research topic: Deep learning methods for aneurysm detection.</li><li>- Summary:<br/>The project focuses on developing deep learning approaches for reliable aneurysm detection from MRA and related modalities. The objectives are to:<ul style="list-style-type: none"><li>* Improve accuracy and robustness of aneurysm detection using advanced neural architectures;</li><li>* Improve weakly annotated datasets for model training, as well as apply data augmentation techniques and multi-modal data analysis to address the data size limitation.</li></ul></li></ul></li></ul>                                                                                      |

- **Postdoctoral researcher** (March 2023–2025), MIND, Inria Paris-Saclay, France.
  - Advisor: [Dr. Bertrand Thirion](#)
  - Co-advisor: [Dr. Demian Wassermann](#)
  - Research topic: Knowledge and representation integration on the brain
  - Summary:
 

The project aims to develop a novel approach for image representation in the context of brain imaging, enabling a more nuanced and flexible assessment of the associations between images and arbitrary queries, unconstrained by traditional “bag of words” limitations. In other words, we want to:

    - \* Provide reliable knowledge from diverse brain studies;
    - \* Address challenges such as lack of statistical power in individual studies, reproducibility, and terminology inconsistency;
    - \* Provide insights into the relationship between brain structure and behavior.
- **Postdoctoral researcher** (August 2021–August 2022), [MLSP-Lab](#), University of Maryland, Baltimore County (UMBC), Maryland, USA.
  - Advisor: [Prof. Tulay Adali](#)
  - Research topics:
    - Dictionary learning for the identification of new interpretable patterns and discriminative features from brain functional network connectivity (FNC) obtained from ICA decomposition of multi-subject resting-state fMRI data for static and dynamic studies.
    - Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, *e.g.*, Schizophrenia.
    - Constrained ICA and IVA for subgroup identification from multisubject fMRI Data.
    - Reproducibility and replicability in neuroimaging data analysis.
- **Research assistant** (2015–2017), DSP-lab, EE Department, Sharif University of Technology, Tehran, Iran.
  - Advisor: [Prof. Massoud Babaie-Zadeh](#)
  - Research topics: Dictionary learning for sparse representation, convex/non-convex optimization.

POSTERS                      “NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature”, [Organization for Human Brain Mapping \(OHBM\)](#), Brisbane, Australia, 2025.

WORKSHOP                    [COGBASES workshop on open science methods for analyzing brain imaging data](#), Paris, France, October 2023.

SUMMER SCHOOL            [PRAIRIE artificial intelligence summer school \(PAISS\)](#), Grenoble, France, July 2018.

TALKS

- **From Contrastive Text-to-Brain Mapping to Multiview Learning of Depressive Disorders through Biological and Discourse Biomarkers**, [MULTISPEECH](#), Inria-Nancy, France, February 2026.
- **Multiview Learning of the Brain: Functional Connectivity, Subgroup Discovery, and Text-to-Brain Mapping**, *IMoPA, ERCN, CHRU de Nancy*, Nancy, France, February 2026.

- **NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation**, *The British Council in Paris*, Paris, France, September 2025.
- **Exploring Brain Function and Structure: From Sparse Coding to Multimodal Meta-Analysis**, *Laboratoire de Physique de l'ENS de Lyon*, Lyon, France, February 2024.
- **Multi-subject fMRI Analysis for Brain Pattern Discovery and Subgroup Identification**, *MIND team, Inria-Saclay*, Paris, France, June 2024.
- **New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity using Dictionary Learning**, *MIND team, Inria-Saclay*, Paris, France, September 2023.
- **Optimal Sensor Placement for Source Extraction**, *MIND team, Inria-Saclay*, Paris, France, June 2023.
- **Optimal Sensor Placement for Source Extraction**, *Diagnostic and Interventional Adaptive Imaging (IADI)*, Nancy, France, January 2023.
- **Optimal Sensor Placement for Source Extraction**, *Centre de Recherche en Automatique de Nancy (CRAN), Department of Biology, Signals and Systems*, Nancy, France, January 2021.

#### Publications

##### PREPRINTS

1. **F. Ghayem**, R. Meudec, J. Dockès, B. Thirion, D. Wassermann, “NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation,” submitted to *Imaging Neuroscience*, July 2025. [\[Paper\]](#) [\[Codes\]](#)
2. R. Meudec, J. Dockès, **F. Ghayem**, D. Wassermann, B. Thirion, “Peaks2Image: Reconstructing fMRI Maps from Stereotactic Coordinates to Enhance Cognitive Meta-Analysis,” submitted to *Imaging Neuroscience*, August 2025. [\[Paper\]](#)

##### REFEREED JOURNAL PUBLICATIONS

1. **F. Ghayem**, B. Rivet, C. Jutten, R. Cabral Farias, “Robust sensor placement for signal extraction”, *IEEE Transactions on Signal Processing*, vol. 69, pp. 4513-4528, 2021.
2. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, “Sparse signal recovery using iterative proximal projection”, *IEEE Transactions on Signal Processing*, vol. 66, no. 4, pp. 879–894, February 2018.

##### CONFERENCE PUBLICATIONS

1. R. Meudec\*, **F. Ghayem\***, J. Dockès, D. Wassermann, B. Thirion, “NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature”, *International Conference on Medical Image Computing And Computer Assisted Intervention (MICCAI)*, October 2024.  
\* [Equal contribution](#)
2. **F. Ghayem**, H. Yang, F. Kantar, S-J. Kim, V. D. Calhoun, T. Adali, “New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity Using Dictionary Learning”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Rhodes island, June 2023.

3. H. Yang, **F. Ghayem**, B. Gabrielson, M. A. B. S. Akhonda, V. D. Calhoun, T. Adali, “Constrained independent component analysis based on entropy bound minimization for subgroup identification from multisubject fMRI data”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, June 2023.
4. H. Yang, MABS. Akhonda, **F. Ghayem**, Q. Long, VD. Calhoun, T Adali, “Independent Vector Analysis Based Subgroup Identification from Multisubject fMRI Data”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2022.
5. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, “Gradient-based algorithm with spatial regularization for optimal sensor placement”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2020.
6. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, “Optimal sensor placement for signal extraction”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2019.
7. M. Sadeghi, **F. Ghayem**, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, “L0Soft:  $\ell_0$  Minimization via Soft Thresholding”, in *Proceedings of the 27th European Signal Processing Conference (EUSIPCO)*, 2-6 September 2019.
8. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, and C. Jutten, “Accelerated dictionary learning for sparse signal representation”, in *13th International Conference on Latent Variable Analysis and Signal Separation, LVA/ICA*, Grenoble, France, 2017.
9. **F. Ghayem** and F. Rassaie, “Helical antenna to measure radiated power density around a BTS; Design and implementation”, in *third Asia-Pacific Conference on Antennas and Propagation (APCAP)*, July 2014.

CO-SUPERVISION I am currently supervising four Master 1 students at Université de Lorraine on a project titled “Enhanced NeuroConText with deep image representation for brain meta-analysis”. Also, I co-supervised three Ph.D. students in collaboration with Prof. Tulay Adali (University of Maryland, Baltimore County, USA), Prof. Jean-Christophe Pesquet (CentraleSupélec, Université Paris-Saclay, France), Prof. Vince D. Calhoun (Translational Research in Neuroimaging and Data Science, USA), and Dr. Seung-Jun Kim (University of Maryland, Baltimore County, USA) on the following projects:

- Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, e.g. Schizophrenia.
- ICA and IVA for subgroup identification from multisubject resting state fMRI Data.
- Reproducibility and replicability in neuroimaging data analysis.

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|----------|-----------------------------------------------------------------------|-----------|
| TEACHING | <b>Signal Processing</b>                                              | 2025-2026 |
|          | - 6 hours lectures + 12 hours practical sessions                      |           |
|          | - Université de Lorraine                                              |           |
|          | <b>Signal Processing</b>                                              | 2024-2025 |
|          | - 6 hours lectures + 12 hours practical sessions                      |           |
|          | - Université de Lorraine                                              |           |
|          | <b>Signals and Systems</b>                                            | 2015      |
|          | - 10 hours practical sessions                                         |           |
|          | - Responsible: Prof. M. Babaei-Zadeh, Sharif University of Technology |           |
|          | <b>Digital Signal Processing II</b>                                   | 2014      |
|          | - 10 hours practical sessions                                         |           |

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|                       | - Responsible: Prof. F. Marvasti, Sharif University of Technology                                                                                                                                                           |                  |
|                       | <b>Signals and Systems</b>                                                                                                                                                                                                  | 2013             |
|                       | - 20 hours practical sessions                                                                                                                                                                                               |                  |
|                       | - Responsible: Dr. M. Derakhtian, Shiraz University                                                                                                                                                                         |                  |
|                       | <b>Electromagnetics</b>                                                                                                                                                                                                     | 2012             |
|                       | - 20 hours practical sessions                                                                                                                                                                                               |                  |
|                       | - Responsible: Dr. M. Derakhtian, Shiraz University                                                                                                                                                                         |                  |
|                       | <b>Electrical Circuit II</b>                                                                                                                                                                                                | 2011             |
|                       | - 20 hours practical sessions                                                                                                                                                                                               |                  |
|                       | - Responsible: Prof. M. A. Masnadi-Shirazi, Shiraz University                                                                                                                                                               |                  |
| QUALIFICATIONS        | I am qualified to serve as Maître de Conférences, Section 26 – Mathématiques appliquées et applications des mathématiques.                                                                                                  |                  |
| HONORS & AWARDS       | - IEEE TMI Distinguished Reviewer Silver Level 2023 – 2024                                                                                                                                                                  | 2024             |
|                       | - Ph.D. scholarship (ranked 2), Université Grenoble Alpes, Grenoble, France.                                                                                                                                                | 2017             |
|                       | - Full travel grant (CHESS project), LVA/ICA workshop, Grenoble, France.                                                                                                                                                    | 2017             |
|                       | - Bronze award in math competition among high school students, Sharif University of Technology.                                                                                                                             | 2008             |
|                       | - Admitted to National Organization for Development of Exceptional Talents (NODET) as high school and pre-university school student.                                                                                        | 2005             |
| COMPUTER SKILLS       | <ul style="list-style-type: none"> <li>• <i>Programming Languages and Softwares:</i> Python, PyTorch, MATLAB</li> <li>• <i>Typesetting:</i> L<sup>A</sup>T<sub>E</sub>X</li> <li>• <i>Toolbox:</i> Nilearn, GIFT</li> </ul> |                  |
| COMMUNITY SERVICES    | <b>Reviewer for the following journals and conferences:</b>                                                                                                                                                                 |                  |
|                       | - AAAI                                                                                                                                                                                                                      | 2026             |
|                       | - ICML                                                                                                                                                                                                                      | 2025             |
|                       | - NeurIPS                                                                                                                                                                                                                   | 2024-2025        |
|                       | - Journal of Machine Learning Research                                                                                                                                                                                      | 2024-2025        |
|                       | - IEEE Transactions on Machine Learning Research                                                                                                                                                                            | 2024-2025        |
|                       | - IEEE Transactions on Medical Imaging                                                                                                                                                                                      | 2023-2026        |
|                       | - IEEE Transactions on Signal Processing                                                                                                                                                                                    | 2019-2022        |
|                       | - IEEE Signal Processing Letters                                                                                                                                                                                            | 2019, 2021, 2023 |
|                       | - International Conf. on Acoustics, Speech, and Signal Proc. (ICASSP)                                                                                                                                                       | 2023             |
|                       | - European Signal Processing Conference (EUSIPCO)                                                                                                                                                                           | 2019-2021        |
|                       | - eNeuro                                                                                                                                                                                                                    | 2022             |
|                       | - Machines                                                                                                                                                                                                                  | 2022-2023        |
| LANGUAGE PROFICIENCY  | - English (Fluent)                                                                                                                                                                                                          |                  |
|                       | - French (Intermediate)                                                                                                                                                                                                     |                  |
|                       | - Persian (Native)                                                                                                                                                                                                          |                  |
| HOBBIES AND INTERESTS | Playing the violin, running, hiking, biking                                                                                                                                                                                 |                  |

## REFERENCES

- Prof. Christian Jutten  
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- Prof. Massoud Babaie-Zadeh  
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- Dr. Bertrand Thirion  
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- Dr. Bertrand Rivet  
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- Dr. Erwan Kerrien  
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- Prof. Tulay Adali  
Adali@umbc.edu